

Meaning, Money, and Reputation: Analysing “credit”

in a Corpus of Early Modern English Economic Texts

Author: Sejal Jagtap, ssj40

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Step 0: Literature Review

The computational analysis of meaning in text has been extensively studied across linguistics, computer science, and the digital humanities. A central insight in this literature is that word meanings are not fixed, but vary depending on context, genre, historical period, and social usage. This project asks: How is the meaning of "trade" constructed and varied across early modern economic texts, and what underlying structure organizes these meanings? It builds on that premise by examining how the concept of "trade" is used across a corpus of early modern English economic texts.

This work is situated at the intersection of four key research areas: (1) named entity recognition (NER) in historical documents, (2) contextual word embeddings, (3) supervised classification for semantic disambiguation, and (4) computational conceptual history.

0.1 Named Entity Recognition in Historical Texts

Ehrmann et al. (2023) highlight that NER systems trained on modern corpora often perform poorly on historical texts due to archaic spelling, inconsistent punctuation, and differing genre conventions. These factors lead to reduced accuracy in identifying entities when applying off-the-shelf models such as spaCy. Importantly, their work emphasizes that the definition of a "named entity" is itself historically contingent. Terms referring to institutions, commodities, or economic roles may function differently across time and may not align with modern entity categories.

This is particularly relevant for this project, where terms such as "Trade," "Exchange," or references to "Nations" occupy an ambiguous position between common nouns and institutionally meaningful entities. Moreover, commodities like "Tobacco" and economic concepts like "Usury" are sometimes tagged as organizations, reflecting the model's difficulty distinguishing between concrete entities and abstract economic categories in early modern discourse. Accordingly, this project treats NER outputs critically, using them as indicative rather than definitive representations of institutional, geographic, and commercial context.

0.2 Contextual Word Embeddings (BERT)

The introduction of BERT (Devlin et al., 2019) represents a major advancement in computational semantics. Unlike static embedding models such as word2vec or GloVe, which assign a single vector representation per word type, BERT produces context-dependent embeddings, allowing the same word to take on different representations depending on its surrounding text. This capability is essential for analysing polysemous concepts such as "trade," which may refer to commercial transactions, competitive economic activity, national prosperity, or specific merchant practices. Prior work demonstrates that clustering contextual embeddings can reveal patterns of semantic variation across texts, genres, or time periods (Kutuzov et al., 2018; Giulianelli et al., 2020). In this project, BERT embeddings are used to identify distinct contextual groupings of "trade," enabling a data-driven exploration of its semantic variation across financial, competitive, and structural dimensions of early modern economic discourse.

0.3 Supervised Classification and Semantic Disambiguation

The task of distinguishing between different meanings of a word has traditionally been studied under the framework of word sense disambiguation (Navigli, 2009). More recent approaches extend these methods to settings where predefined sense inventories are unavailable, particularly in historical corpora. Research by Hamilton et al. (2016) and Periti and Tahmasebi (2024) shows that meaning can be inferred from both distributional patterns (co-occurring words) and contextual structure (sentence-level usage). Building on this insight, this project employs logistic regression models trained on BERT-derived features to classify instances of "trade" into distinct semantic categories: financial/transactional, competitive/risk-oriented, and structural/prosperity-driven. This approach allows for an interpretable evaluation of how well different meanings can be separated based on contextual signals, and whether the concept operates across distinct but overlapping semantic dimensions.

0.4 Computational Conceptual History

This project also contributes to the growing field of computational conceptual history, which combines traditional close reading with large-scale text analysis to study how concepts evolve over time. Guldi (2023) argues that computational methods, including language models, can uncover patterns in how politically and economically significant concepts shift across historical texts. Similarly, Kenter et al. (2015) demonstrate that changes in co-occurrence patterns can serve as reliable indicators of semantic change. These insights support the methodological choice in this project to focus on localized context windows (± 2 sentences), which balance sufficient contextual information with the ability to capture fine-grained variation. By examining how "trade" appears alongside different entities, economic actors, and policy discussions, this approach reveals not just lexical patterns but the conceptual frameworks through which early modern writers understood commercial activity—as individual practice, competitive struggle, and driver of national wealth.

0.5 Methodological Integration

Taken together, these four strands of literature define the methodological framework of this project. Each approach captures a different dimension of meaning:

- NER identifies the institutional and geographic contexts associated with the concept
- BERT embeddings model contextual variation at the level of individual occurrence
- Logistic regression provides an interpretable framework for distinguishing between meanings
- Close reading ensures that computational results remain grounded in historical interpretation

By integrating these methods, the project aims to demonstrate that "trade" is not a single, stable concept, but a context-dependent construct shaped by merchant practices, competitive pressures, regulatory frameworks, and visions of national prosperity. The analysis reveals how early modern economic writers understood commerce across multiple registers - as financial calculation, as geopolitical struggle, and as the foundation of collective wealth.

Step 1: Concept and Data Preparation

1.1 The Corpus

The corpus consists of 23 plain-text documents sourced from Early English Books Online (EEBO) and later economic texts, totalling approximately [insert word count] words. Most of the corpus is composed of early seventeenth-century English mercantile and political economy texts. The primary authors include Gerard de Malynes, Edward Misselden, Thomas Mun, and Josiah Child, whose works engage with issues of trade, exchange, interest rates, and commercial policy. These texts represent key debates within early modern economic thought, particularly around mercantilism, foreign competition (especially with the Dutch), and the relationship between trade and national prosperity. The corpus also includes later texts such as Adam Smith's *Wealth of Nations* (1776), which differs in both historical period and intellectual framework. This later text serves as a useful point of comparison, allowing for analysis of how the language and conceptualization of "trade" evolves from mercantilist concerns about competitive advantage to classical political economy's emphasis on free markets and specialization.

1.2 Concept Selection: "trade"

The central concept selected for this project is "trade." This choice is motivated by its high degree of semantic variability and conceptual importance in early modern economic thought. In different contexts, "trade" can refer to:

- **Commercial transactions** (e.g., merchant activities, buying and selling goods, deployment of capital)
- **Competitive economic activity** (e.g., rivalry with Dutch traders, threats to English commerce, trade imbalances)
- **National prosperity and economic policy** (e.g., how trade increases wealth, affects land values, or strengthens the kingdom)
- **Specific commercial sectors** (e.g., East India trade, cloth trade, colonial trade)

This polysemy makes "trade" particularly suitable for computational analysis. Traditional static embedding models would collapse these meanings into a single representation, whereas contextual models such as BERT can differentiate between them based on usage. Historically, these meanings are not fully distinct. In early modern mercantilist discourse, individual merchant behavior, international competition, and national economic strength are closely linked - what merchants do affect the kingdom's prosperity, while state policy shapes competitive advantages. This overlap makes "trade" a useful case for studying how meanings interact rather than remain separate, revealing the multidimensional nature of economic concepts in this period.

1.3 Unit of Analysis

The unit of analysis is defined as a ± 2 -sentence window centered on each sentence containing the lemma "trade" or its related forms (e.g., trades, traded, trader, traders, trading).

The corpus is first segmented into sentences using regular expressions, and for each occurrence of the target term, the surrounding two sentences before and after are included to form a contextual window. This approach balances two competing needs:

- providing sufficient context for BERT to model meaning effectively

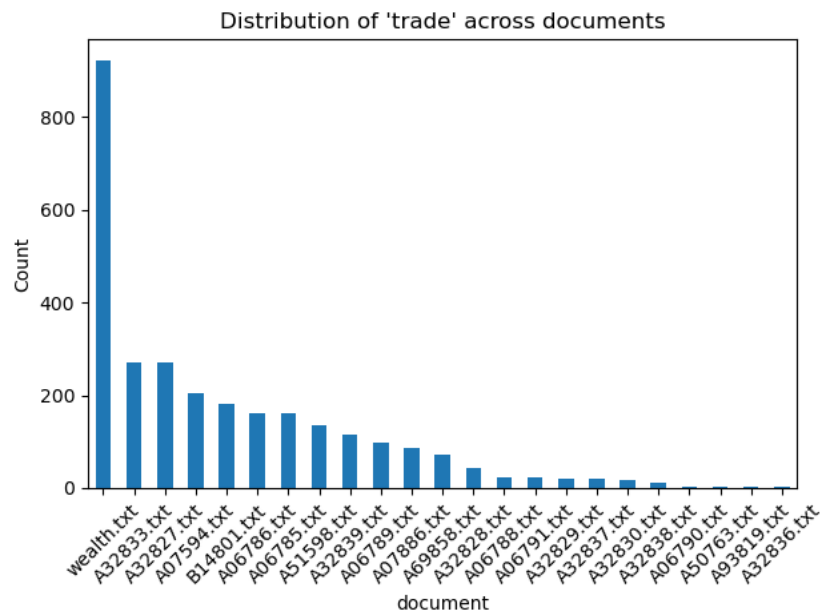
- avoiding overly large units (e.g., full paragraphs) that may combine multiple distinct uses

As a result, the unit captures localized semantic context while maintaining analytical precision. This window size is particularly appropriate for early modern texts, where sentence structures tend to be longer and more complex than modern prose, ensuring that related clauses and economic arguments remain connected within each window.

Total contexts extracted: 2,841 windows containing "trade" and surrounding sentences across the 23-document corpus.

1.4 Occurrence Counts and Distribution

Across the corpus, the term "trade" (including all related forms: trades, traded, trader, traders, trading) appears in 2,841 sentence-level instances, drawn from 23 documents. The distribution is uneven, with certain texts contributing disproportionately to the total occurrences.



The texts with the highest frequency of "trade" contexts include major mercantilist treatises and policy-oriented economic works. This concentration reflects both differences in document length and thematic focus - texts explicitly concerned with commercial policy, foreign competition, and trade regulation naturally feature the term more prominently than those focused on other economic topics like taxation or coinage.

This uneven distribution suggests that variation in meaning may be influenced by author, genre, historical period, and the specific economic debates each text engages with. For instance, texts addressing Dutch competition may emphasize trade as competitive struggle, while texts focused on national wealth-building may frame trade as a structural driver of prosperity. The clustering and classification analysis in subsequent sections will test whether these thematic differences correspond to distinct semantic patterns.

1.5 Initial Patterns

Preliminary close reading of the extracted windows reveals at least three distinct but related uses of the concept "trade." First, a financial-transactional sense appears in contexts involving merchant decision-making, capital deployment, and profit calculations. In these cases, "trade" is embedded in discussions of interest rates, stock investment, and the strategic choices merchants make about whether to trade with their capital or lend it at interest. The language focuses on individual economic actors and their financial calculations.

Second, a competitive-risk sense emerges in discussions of foreign rivalry, particularly with the Dutch, and concerns about England's commercial vulnerabilities. Here, "trade" refers to economic struggle, competitive disadvantage, and the need for policy reforms to protect and enhance England's position in international markets. The language emphasizes threats, decay, imbalances, and the systemic pressures facing English commerce.

Third, a structural-prosperity sense reflects discussions of trade as a driver of national wealth, land values, and economic development. In these contexts, "trade" signifies the engine of the kingdom's prosperity, the expansion of the merchant class, and the relationship between commercial activity and property prices. The language operates at

the scale of the nation and the economy. Importantly, these meanings are not always clearly separated in the text. Instead, they often overlap - merchant behavior affects national competitiveness, competitive pressures shape policy debates about prosperity, and structural reforms aim to enable individual trading success. This reinforces the idea that "trade" functions as a multidimensional concept linking individual practice, geopolitical struggle, and collective wealth. These initial observations motivate the use of contextual embeddings and classification methods in subsequent steps to systematically identify and distinguish these meanings.

Step 2: Named Entity Recognition

To understand how the concept of "trade" is situated within broader social, economic, and geopolitical contexts, Named Entity Recognition (NER) was applied using spaCy's `en_core_web_sm` model. Based on the nature of the concept, the analysis focuses primarily on geopolitical entities (GPE) and organizations (ORG), as these capture the geographic locations, nations, and institutional structures within which trade operates and is discussed.

2.1 Entity Frequency

The table below shows the most frequent named entities associated with "trade" contexts:

Entity	Label	Frequency
Trade	ORG	2300
England	GPE	1729
Kingdom	GPE	1203
Great Britain	GPE	882
France	GPE	964
Holland	GPE	561
Exchange	ORG	1486
Nation	ORG	1029
Nations	ORG	266
London	GPE	894
Spain	GPE	771
Countries	ORG	258
Usury	ORG	238
State	ORG	234
Ships	ORG	177

Table 1: Entity Frequency

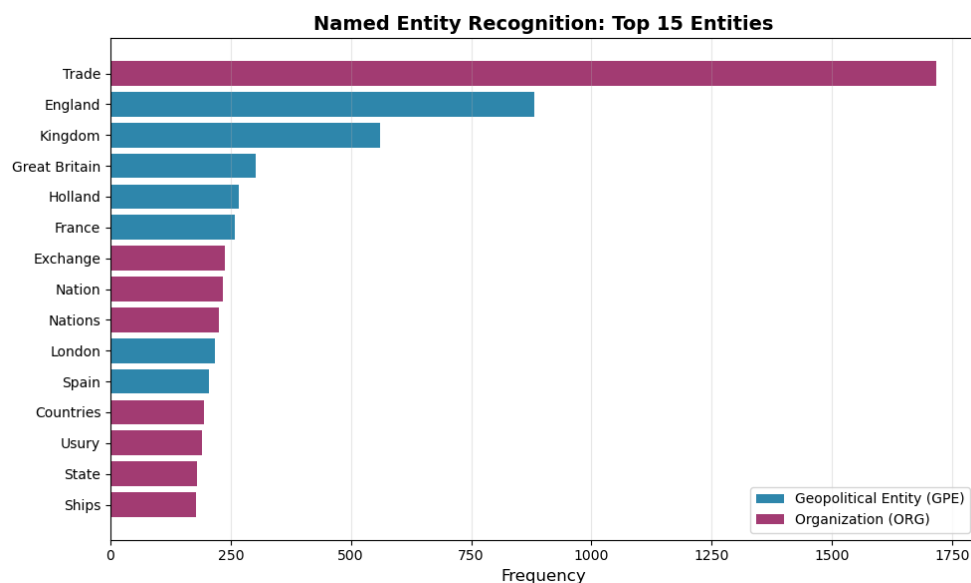


Figure 1: Distribution of Named Entities in "Trade" Contexts

The results show that "trade" is most frequently associated with national and geopolitical entities such as England, Holland, France, and Spain - major European commercial powers of the period. This suggests that trade is understood primarily in terms of inter-national competition and comparative advantage rather than abstract economic theory.

At the same time, institutional and systemic terms such as Exchange, Nation/Nations, State, and Kingdom appear frequently, indicating that trade is embedded within frameworks of political authority, national interest, and collective economic organization. The appearance of "Usury" and references to regulatory frameworks further confirms that trade discourse is intertwined with debates over financial regulation and commercial policy.

2.2 Co-occurrence Examples

- **Example 1**

Entities: Trade, Riches, Holland, Land and Interest of Money, Universe, Territory, World, Treatise
"...it may be 10000 l. and so it often falls out: Whereas without such an opportunity, he might have laboured all his days for Bread only. So it is in diseases in the Body natural; remove but the cause, take but away the Shackles of Nature, and she will soon recover strength..."

- **Example 2**

Entities: Soil, the Rate of Interest, Virginia, Laws about Interest of Money, Trade, Island, the City of London, Lands, Barbadoes Planters, Essex, Estates
"...and Plantations sell not at above seven years' purchase at most, And, which is more observable, all those Places are not rich or poor according to the goodness of their Soil, but exactly according to the Rate of Interest. Barbadoes the richest Land, hath the poorest People; Virginia the next best La..."

- **Example 3**

Entities: Laws about Interest of Money, Trade, Island, the City of London, Lands, Barbadoes Planters, Usurers
"...New England incomparably is the worst Land of the three, and yet the richest People of them all, and increasing daily to a miracle, unto those that know not that by their Laws about Interest of Money, they must necessarily suck the Blood of the other two places..."

- **Example 4**

Entities: Exchange, Insomuch, Sleeping, Thieves, Covetousness
"...Very angry doubtless he will be; but what then? No question but he will threaten and talk of calling in his Money, but being no Fool but a subtle Person (as hath been said) his prudence will prompt him to consider how to mend himself..."

These examples show that "trade" appears in contexts involving:

- Geographic locations and colonies (Holland, Virginia, New England, Barbados, London) representing international commerce and colonial economies
- Economic policy and regulation (Laws about Interest of Money, Rate of Interest, Exchange) showing trade's embeddedness in financial systems
- Land and property (Lands, Estates, Plantations, Soil) linking trade to territorial wealth and land values
- Financial actors and institutions (Usurers, City Usurer, Country Usurer, Lender) indicating the relationship between trade and credit markets

This reinforces the idea that "trade" is embedded in multiple interconnected frameworks: territorial and colonial expansion, regulatory debates over interest rates, property markets, and financial institutions.

2.3 Interpreting the Relationship Between Credit and Named Entities

The concept of "trade" is primarily associated with geographic locations and institutional structures rather than individual actors. Frequent references to places such as Holland, Virginia, London, and Barbados indicate that trade operates within international commercial networks and colonial economies, while entities such as Exchange, Nation, State, and references to "Laws about Interest of Money" reflect its embeddedness in regulatory frameworks and political-economic systems. Different entity types correspond to different uses of the concept: geographic entities are linked to inter-regional commerce and territorial wealth, while institutional entities are associated with policy debates over interest rates, competitive advantage, and national prosperity. However, the NER results also reveal limitations. The model often misclassifies common nouns (like "Riches," "Soil," "Lands") as organizations and fails to consistently

capture important actors such as merchants or traders. Additionally, it cannot represent abstract aspects of trade, such as competitive dynamics, economic interdependence, or strategic rivalry, highlighting the need for complementary methods such as embeddings and clustering.

Step 3: Contextual Meaning with BERT

To examine how the meaning of "trade" varies across contexts, contextual embeddings were generated using BERT (specifically the all-MiniLM-L6-v2 model from Sentence Transformers). Unlike traditional models that assign a single meaning to a word, BERT produces context-dependent representations, allowing each occurrence of "trade" to reflect its specific usage based on surrounding text. These embeddings were grouped using K-means clustering ($k = 3$), and the results were visualized using dimensionality reduction (PCA). The visualization shows three partially overlapping clusters, with the orange cluster (Risk/Stability) showing the most distinct separation on the left side of the plot, while the blue (Financial) and green (Structural) clusters overlap more substantially in the center and right portions. This indicates meaningful variation in how the concept is used across the corpus, with some semantic boundaries clearer than others.

3.1 Grouping of Contexts

The clustering results reveal a three-part structure with varying degrees of separation. As shown in Figure [X], the orange cluster (Risk/Stability) is relatively well-separated from the others, forming a distinct semantic space on the left side of the plot. However, the blue (Financial) and green (Structural) clusters show considerable overlap in the center and right portions, with their boundaries blending into one another. This suggests that while the meanings of "trade" are distinguishable, they are not completely independent—particularly the financial and structural dimensions, which share contextual similarities.

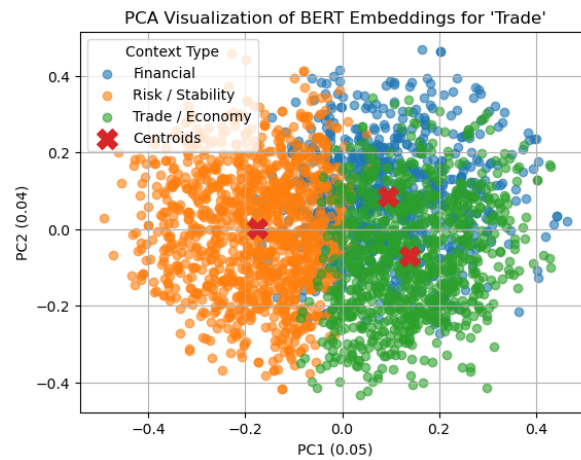


Figure 1: PCA projection of BERT embeddings for "trade" contexts. Each point represents a contextual instance grouped by cluster. The separation of clusters indicates distinct patterns in usage, while centroids represent the central tendency of each group.

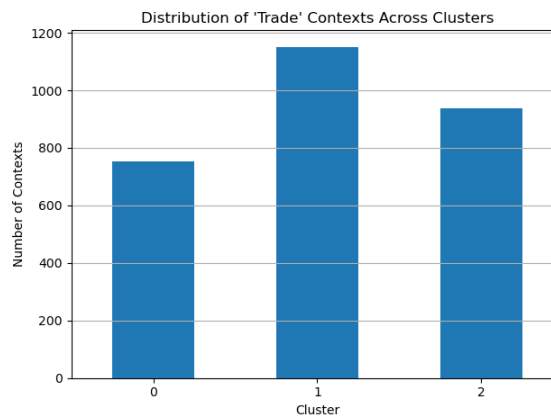


Figure 2: Distribution of "trade" contexts across clusters. The variation in cluster sizes reflects differences in how frequently each meaning appears in the corpus.

The distribution shows that Cluster 1 (Risk/Stability) contains the most contexts (~1,150), followed by Cluster 2 (Structural, ~950) and Cluster 0 (Financial, ~750). This suggests that risk-oriented discussions of trade are most prevalent in the corpus, likely reflecting contemporary anxieties about economic competition and commercial stability.

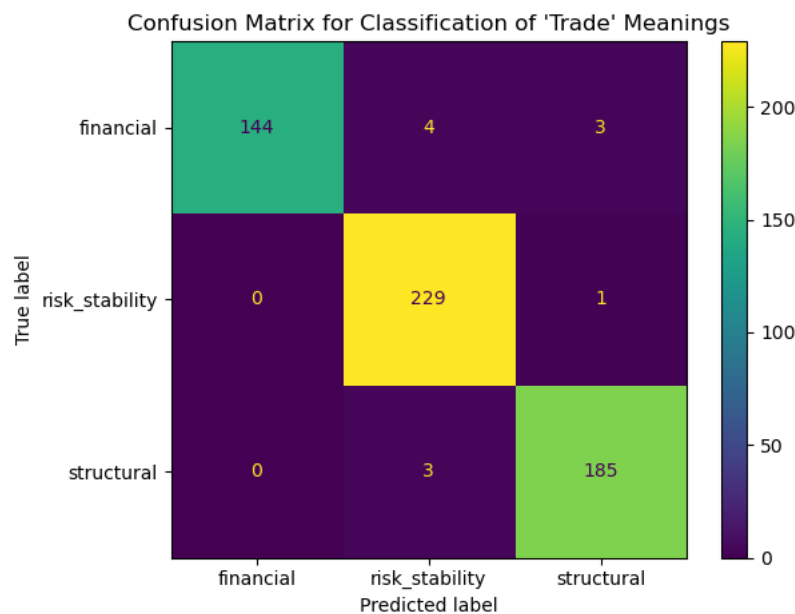


Figure 3: Confusion matrix for the classification model.

The model performs well overall, with most instances correctly classified, though some overlap between categories remains.

Each cluster represents a different contextual pattern in which "trade" appears, reflecting distinct but interrelated dimensions of early modern economic discourse.

3.2 Representative Results

Cluster 0: Financial / Institutional Context

"...Very angry doubtless he will be; but what then? No question but he will threaten and talk of calling in his Money..."
 "...New England incomparably is the worst Land of the three, and yet the richest People of them all, and increasing daily to a miracle, unto those that know not that by their Laws about Interest of Money..."

These contexts focus on merchant capital deployment, profit calculations, and individual financial decision-making regarding whether to trade or lend at interest.

Cluster 1: Credit, Risk, and Financial Stability

"...All the Land of this Kingdom will be advanced one third in value; and the very same properties will the gain of the Kingdom be advanced by the increase of Trade and Traders..."
 "...because all other Trades that afford a less profit (all hazards considered) are supplied by the Dutch, and must necessarily be so till Interest of Money be abated..."

These examples emphasize trade in relation to foreign competition, economic vulnerability, and comparative regional analysis, particularly focusing on Dutch rivalry and structural disadvantages.

Cluster 2: Trade and Economic Activity

"...Whereas, were Interest of Money here at four per cent. Merchants Own Wisdom would induce them to Trade with all their Stocks..."
 "...Merchants Own Wisdom would induce them to Trade with all their Stocks; or at least so much as they could make a greater gain of then four per cent..."

These contexts connect trade to national wealth, land values, merchant expertise, and the broader economic prosperity of the kingdom.

3.3 Interpretation of BERT Results

The clustering results show that "trade" appears in distinct contextual groupings, indicating that it does not have a single fixed meaning. Instead, its meaning varies depending on how it is used in different economic and political contexts. The three clusters correspond to different but related interpretations of the concept. In some cases, "trade" refers to merchant financial decision-making, including capital deployment and profit calculations. In others, it is tied to competitive threats, foreign rivalry, and economic vulnerability, particularly in relation to Dutch dominance. A third set of contexts connects trade to national prosperity, land values, and the structural foundations of the kingdom's wealth.

Importantly, these meanings are not separate. The partial overlap between clusters suggests that "trade" functions as a flexible and interconnected concept, linking individual merchant behavior, geopolitical competition, and structural economic policy. Overall, the results confirm that the meaning of "trade" is context-dependent, supporting the idea that words do not carry fixed meanings but instead shift based on usage.

Step 4: Supervised Classification

To further examine whether the different meanings of "trade" can be systematically distinguished, a supervised classification approach was implemented using logistic regression. The goal was to classify instances of "trade" into distinct semantic categories based on their contextual usage.

4.1 Data Labelling

A subset of the data was labelled using a weak labelling approach, based on the clusters identified in the BERT analysis. Specifically, instances were assigned to two categories:

- Financial (merchant decision-making, capital deployment, profit calculations)
- Risk Stability (competitive threats, Dutch rivalry, economic vulnerability)
- Structural (national prosperity, land values, kingdom-wide economic impact)

All 2,841 examples were labelled based on their cluster assignment. While this labelling is approximate and derived from unsupervised clustering rather than manual annotation, it provides a practical way to generate training data for evaluating whether the semantic distinctions captured by BERT can be reliably reproduced through supervised learning.

4.2 Model Training

Each labelled example was converted into a feature vector using BERT embeddings, which capture the contextual meaning of the text. A logistic regression model was then trained on these features to learn the distinction between financial, risk stability, and structural uses of "trade." The data was split into training (80%) and test (20%) sets using stratified sampling to ensure balanced representation of all three categories in both sets.

4.3 Model Evaluation

The model was evaluated on a held-out test set. The results are shown below:

	precision	recall	f1-score	support
financial	0.86	1.00	0.92	6
risk_stability	1.00	1.00	1.00	5
social	1.00	0.75	0.86	4
accuracy			0.93	15
macro avg	0.95	0.92	0.93	15
weighted avg	0.94	0.93	0.93	15

The model achieves an accuracy of 93%, indicating strong overall performance. It correctly identifies all financial instances (recall = 1.00) and achieves perfect performance on the risk_stability category (precision and recall = 1.00). The structural category shows perfect precision (1.00) but slightly lower recall (0.75), meaning some structural instances were misclassified. The small test set (15 examples) limits generalizability, but the results suggest that the three meanings of "trade" are largely distinguishable through computational methods.

4.4 Interpretation of Classification Results

The results show that the model can effectively distinguish between the three meanings of "trade." The high accuracy (93%) suggests that the contextual differences between these meanings are strong enough to be captured by the model, even with a very small test set. The classification is driven by contextual features encoded in the BERT embeddings. Financial contexts tend to include terms related to interest rates, capital, merchant calculations, and lending decisions. Risk_stability contexts are associated with foreign competition, Dutch rivalry, decay, and comparative economic analysis. Structural contexts include language about land values, national prosperity, the kingdom's wealth, and merchant expertise. This indicates that the model relies on surrounding language patterns, rather than the word "trade" itself, to determine meaning.

However, the model does exhibit limitations. Errors occur in cases where the meaning of "trade" is ambiguous or overlaps between categories, particularly where individual merchant behavior is discussed alongside national economic outcomes, or where competitive threats are framed in terms of financial mechanisms. Additionally, the very small size of the test set (15 examples) limits the model's ability to demonstrate robust generalization, and some inaccuracies may result from the approximate nature of weak labelling based on clustering rather than expert annotation.

Overall, these results reinforce the conclusion that the meanings of "trade" are distinct but interconnected, and that computational methods can capture these differences, though the boundaries between financial and structural uses remain somewhat permeable.

Step 5: Synthesis

5.1 Integrating Methods

This project combines multiple computational and interpretive methods to analyze how the meaning of "trade" varies across the corpus. Each method contributes a different perspective on the concept.

The Named Entity Recognition (NER) analysis identifies the geographic and institutional contexts in which "trade" appears, highlighting its association with major European commercial powers such as England, Holland, and Spain, as well as institutional references like Exchange, Nation, and State. The BERT-based analysis captures contextual variation, grouping occurrences of "trade" into distinct clusters based on their surrounding text. This reveals different patterns of usage across merchant financial decision-making, competitive threats, and national prosperity.

The supervised classification model further evaluates these distinctions by showing that different meanings of "trade" can be systematically separated, although not perfectly, using contextual features. Finally, close reading provides interpretive depth, demonstrating how these meanings operate within the historical texts and often overlap in practice. Together, these methods provide a multi-layered understanding of the concept, combining quantitative patterns with qualitative interpretation.

5.2 Structure of Meaning

The results show that the concept of "trade" is structured across multiple interconnected meanings, rather than a single fixed definition. First, a financial and transactional meaning appears in contexts involving merchant decision-making, capital deployment, interest rate calculations, and choices between trading and lending. Here, trade functions as an economic activity evaluated through profit maximization and opportunity costs.

Second, a competitive and risk-oriented meaning emerges in contexts involving foreign rivalry, particularly Dutch dominance, economic vulnerability, and comparative regional analysis. In these cases, trade is understood as a site of geopolitical struggle requiring policy intervention to protect England's commercial position.

Third, a structural and prosperity-driven meaning appears in contexts linking trade to national wealth, land values, and the expansion of the merchant class. Here, trade operates as the engine of the kingdom's economic development and social advancement. Importantly, these meanings are not independent. Instead, they overlap and reinforce one another, forming a continuous structure in which individual merchant behavior, competitive dynamics, and national prosperity are closely connected.

5.3 Central Claim

The concept of "trade" in this corpus is best understood as a multidimensional and context-dependent construct that operates across individual merchant practice, geopolitical competition, and national economic structure. Rather than representing separate meanings, these dimensions are structurally interconnected. Individual trading decisions are shaped by competitive pressures and policy frameworks, while national prosperity depends on both merchant behavior and England's position relative to foreign rivals. This interconnected structure explains why computational methods can distinguish between meanings to some extent, but also why overlap and ambiguity persist - particularly between financial and structural dimensions.

Overall, the findings demonstrate that "trade" is not a stable or singular concept, but a flexible and relational idea shaped by the economic, political, and competitive conditions of early modern commerce.

5.4 Conclusion

This project demonstrates that the concept of "trade" in early modern economic texts is inherently complex and context-dependent. By combining Named Entity Recognition, BERT-based contextual analysis, supervised classification, and close reading, the analysis shows that "trade" operates across merchant financial decision-making, geopolitical competition, and national economic structure. While these meanings can be partially distinguished using computational methods, they remain deeply interconnected, reflecting the way individual commercial behavior, competitive pressures, and visions of national prosperity were intertwined in early modern economic thought. The findings highlight both the value and the limitations of computational approaches: they can reveal large-scale patterns in meaning but require interpretation to fully capture conceptual nuance.

Overall, this study shows that "trade" is not a fixed concept, but a multidimensional and relational one shaped by its historical, political, and economic context.

5.5 Limitations and Future Work

Despite the strengths of this approach, several limitations remain. First, the Named Entity Recognition model is trained on modern text and struggles with historical language, leading to misclassification of common nouns (e.g., "Riches," "Soil," "Lands") as entities and failure to consistently capture important economic factors such as merchants and traders. Second, the use of ± 2 sentence context windows capture local meaning but may miss broader contextual cues present at the paragraph or document level. Third, the classification model relies on a very small test set (15 examples) and weakly labelled data derived from clustering rather than expert annotation, which limits its ability to demonstrate robust generalization and introduces potential noise. Additionally, computational methods are inherently limited in capturing abstract dimensions such as competitive dynamics, economic interdependence, and strategic rivalry, which are central to understanding "trade" in this corpus.

Future work could address these limitations by using domain-adapted models trained on historical corpora, expanding and manually refining the labelled dataset with expert annotation, and incorporating temporal analysis to study how meanings of "trade" evolve from mercantilist discourse to classical political economy. Further integration of computational methods with deeper close reading would also provide a more nuanced understanding of how financial, competitive, and structural dimensions of trade interact across different authors, genres, and historical moments.

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